



# KTU NOTES

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**KTU STUDY MATERIALS | SYLLABUS | LIVE  
NOTIFICATIONS | SOLVED QUESTION PAPER**

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Soln:

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 3 & -1 \\ 5 & 3 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & 4 & -1 \\ 0 & 8 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & 4 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$\therefore$  Rank of  $A = 2$

2. What kind of conic section is represented by the quadratic form  $7x_1^2 + 6x_1x_2 + 7x_2^2 = 20$  and convert it into canonical form.

Soln:

$$A = \begin{bmatrix} 7 & 3 \\ 3 & 7 \end{bmatrix}$$

Characteristic equation is  $\lambda^2 - \text{trace } A \lambda + \det A = 0$   
 $\lambda^2 - 14\lambda + 40 = 0$   
 $(\lambda - 10)(\lambda - 4) = 0$

$\therefore$  Eigen values are  $\lambda = 4, 10$

Canonical form is  $Q = \lambda_1 y_1^2 + \lambda_2 y_2^2$   
 $Q = 4y_1^2 + 10y_2^2$

$$\frac{y_1^2}{5} + \frac{y_2^2}{2}$$

$$\begin{bmatrix} 0 & -1 & 5 & -8 \\ 0 & -6 & 5 & -4 \\ 0 & -3 & 2 & -4 \end{bmatrix}$$

$R_4 \rightarrow$

$$\sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -7 & 5 & -8 \\ 0 & 0 & 5/7 & 20/7 \\ 0 & 0 & -1/7 & -4/7 \end{bmatrix}$$

$R_4 \rightarrow$

$$\sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -7 & 5 & -8 \\ 0 & 0 & 5/7 & 20/7 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\therefore \text{Rank } A = \text{Rank } \tilde{A} = \text{no. of unknowns}$

$\therefore$  System of equations have unique

By Back substitution,

$$\frac{5}{7}z = \frac{20}{7} \Rightarrow z = 4$$

$$-7y + 5z = -8 \Rightarrow 7y - 8 =$$

$$x + 2y - z = 3 \Rightarrow x + 2 \times 4 -$$

$$\Rightarrow x + 4 = 3$$

$\therefore$  solution is  $x = -1, y = 4, z = 4$

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Eigen vector corresponding to  $\lambda = 1$

$$\text{Consider } (A - \lambda I)x = 0$$

$$(A - I)x = 0$$

$$A - I = \begin{bmatrix} 1 & 1 & 2 \\ -1 & 2 & 1 \\ 0 & 1 & 3 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} -1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix}$$

$$\sim \begin{bmatrix} -1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \quad R_2$$

$$\text{rank} = 2$$

$\therefore$  no. of eigen vectors corresponding to

is no. of variables - rank =

By Back substitution  $x_2 + 2x_3 = 0$

$$\text{Let } x_3 = 1$$

$$-x_1 + x_2 + x_3 = 0$$

$$x_1 - x_2 + x_3 = 0$$

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$$\sim \begin{bmatrix} -1 & 1 & 2 \\ 0 & -1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \quad R_3 \rightarrow R_3 +$$

$$\text{Rank} = 2$$

$$\text{no. of Eigen vectors} = 3 - 2 = 1$$

By Back substitution,

$$-x_2 - x_3 = 0 \Rightarrow x_2 = -$$

$$\text{Let } x_3 = 1 \therefore x_2 = -1$$

$$-x_1 + x_2 + 2x_3 = 0$$

$$x_1 = x_2 + 2x_3 = -1 + 2$$

Eigen vector corresponding to  $\lambda = 2$

$$X_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

Eigen vector corresponding to  $\lambda = 3$

$$\text{Consider } (A - 3I)X = 0$$

$$A - 3I = \begin{bmatrix} -2 & 1 & 2 \\ 1 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

By Back substitution,

$$-\frac{3}{2}x_2 = 0 \Rightarrow x_2 = 0$$

$$-2x_1 + x_2 + 2x_3 = 0$$

$$\text{Let } x_3 = 1$$

$$2x_1 = x_2 + 2x_3 = 0 + 2 = 2$$

$$x_1 = 1$$

Eigen vector corresponding to  $\lambda = 3$

$$X_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

12 a) For what value of  $a$  and  $b$  the equations  $x+y+z=6$ ,  $x+2y+3z=10$  have 1) no solution 2) unique solution 3) one solution.

$$\begin{bmatrix} 0 & 0 & a-3 & b-10 \end{bmatrix}$$

No solution when  $\text{Rank } A \neq \text{Rank } B$

when  $a=3$  and  $b \neq 10$  ,  $\text{Rank } A = 2$  ,  $\text{Rank } B = 1$

$\therefore$  No solution when  $a=3$  ,  $b \neq 10$

Unique solution when  $\text{Rank } A = \text{Rank } B = 3$   
i.e. when  $a \neq 3$  and  $b = 10$

Infinite no. of solutions when  $\text{Rank } A = \text{Rank } B = 2$   
i.e. when  $a=3$  and  $b=10$

$\therefore \text{Rank } A = \text{Rank } B = 2$

12 b) Find the matrix of transformation that  
matrix  $A =$

$$\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

diagonal matrix.



$$\sim \begin{bmatrix} 8 & -6 & 2 \\ 0 & \frac{20}{8} & -\frac{10}{4} \\ 0 & -\frac{10}{4} & \frac{5}{2} \end{bmatrix} \quad \begin{matrix} R_2 \\ R_3 \end{matrix}$$

$$\sim \begin{bmatrix} 8 & -6 & 2 \\ 0 & \frac{20}{8} & -\frac{10}{4} \\ 0 & 0 & 0 \end{bmatrix} \quad R_3$$

By Back substitution  $\frac{20}{8}x_2 - \frac{10}{4}x_3 = 0$

$$x_2 = x_3$$

$$\text{Let } x_2 = 2 \Rightarrow x_3 = 2$$

$$8x_1 - 6x_2 + 2x_3 = 0$$

$$8x_1 = 6x_2 - 2x_3 = 8x_2 - 2x_3$$

Eigen vector corresponding to  $\lambda = 1$

$$X_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$$\sim \begin{bmatrix} 0 & -16/5 & -8/5 \\ 0 & -8/5 & -4/5 \end{bmatrix}$$

$$\sim \begin{bmatrix} 5 & -6 & 2 \\ 0 & -16/5 & -8/5 \\ 0 & 0 & 0 \end{bmatrix}$$

By Back substitution  $-\frac{16}{5}$   
 $-2x_3$

$$\text{Let } x_2 = 1$$

$$x_3 = -2$$

$$5x_1 - 6x_2 + 2x_3 = 0$$

$$5x_1 = 6x_2 - 2x_3 =$$

$$x_1 = 2$$

Eigen vector corresponding to

$$x_2 = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$$

By Back substitution,

$$-\frac{20}{7}x_2 - \frac{40}{7}x_3 = 0$$

$$-x_2 - 2x_3 = 0 \Rightarrow$$

$$\text{Let } x_3 = 1$$

$$x_2 = -2$$

$$-7x_1 - 6x_2 + 2x_3 = 0$$

$$7x_1 = -6x_2 + 2x_3 =$$

$$x_1 = 2$$

Eigen vector corresponding to  $\lambda = -1 =$

$$X_3 = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$$

$\therefore$  matrix of transformation that diagonalizes  $A$  is

$$P = \begin{bmatrix} 1 & 2 & 2 \\ 2 & -2 & -2 \\ 2 & -2 & 1 \end{bmatrix}$$

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83. Find the derivative of  $w = x^2 + y^2$  w  
the path  $x = at^2$ ,  $y = 2at$ .

$$\begin{aligned}\frac{dw}{dt} &= \frac{\partial w}{\partial x} \times \frac{dx}{dt} + \frac{\partial w}{\partial y} \times \frac{dy}{dt} \\&= 2x \times 2at + 2y \times 2a \\&= 2at^2 \times 2at + 2 \times 2at + 2a \\&= \underline{\underline{4a^2t^3 + 8a^2t}}\end{aligned}$$

$$\frac{\partial z}{\partial y} = \frac{1}{2\sqrt{3x+2y}} = \frac{1}{\sqrt{3x+2y}}$$

$$\left(\frac{\partial z}{\partial y}\right)_{(2,5)} = \frac{1}{\sqrt{3(2)+2(5)}} = \frac{1}{\sqrt{6+10}}$$

$$= \frac{1}{\sqrt{16}} = \frac{1}{4} //$$

Slope of the surface in y-direction

$$x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y} + z \frac{\partial}{\partial z}$$

a) Let  $a = \frac{x}{y}$ ,  $b = \frac{y}{z}$ ,  $c = \frac{z}{x}$

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial u}{\partial a} \times \frac{\partial a}{\partial x} + \frac{\partial u}{\partial c} \times \frac{\partial c}{\partial x} \\ &= \frac{\partial u}{\partial a} \times \frac{1}{y} + \frac{\partial u}{\partial c} \times \frac{-z}{x^2} \\ &= \frac{\partial u}{\partial a} \times \frac{1}{y} - \frac{\partial u}{\partial c} \times \frac{z}{x^2} \end{aligned}$$



$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial b} \times \frac{\partial b}{\partial z} + \frac{\partial u}{\partial c} \times \frac{\partial c}{\partial z}$$

$$= -\frac{y}{z^2} \frac{\partial u}{\partial b} + \frac{\partial u}{\partial c} \times \frac{1}{x}$$

$$x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y} + z \cdot \frac{\partial u}{\partial z}$$

$$= 0$$

$$\text{ie, } \left[ x \frac{\partial u}{\partial a} \times \frac{1}{y} - \frac{\partial u}{\partial c} \times \frac{z}{x} \right] + \left[ -\frac{\partial u}{\partial a} \right]$$

$$+ \left[ -\frac{y}{z} \frac{\partial u}{\partial b} + \frac{\partial u}{\partial c} \frac{z}{x} \right]$$

$$= \underline{\underline{0}}$$

$$f_x = y \quad ; \quad f_x(x_0, y_0, z_0) = y_0$$

$$f_y = x \quad ; \quad f_y(x_0, y_0, z_0) = x_0$$

$$f_z = 2z \quad ; \quad f_z(x_0, y_0, z_0) = 2z_0$$

$$L(x, y, z) = x_0 y_0 + z_0^2 + y_0(x - x_0) + x_0(y - y_0)$$

$$y + 2z - x = x_0 y_0 + z_0^2 + y_0 x - y_0 x_0 + x_0 y -$$

$$y + 2z - x = \cancel{y_0 x} \quad x_0 y + 2z_0 z + y_0 x$$

Comparing coefficients of  $x, y, z$ .

$$y = x_0 y \quad \Rightarrow \quad x_0 = 1$$

$$2z = 2z_0 z \quad \Rightarrow \quad z_0 = 1$$

$$-x = y_0 x \quad \Rightarrow \quad y_0 = -1$$

$$\therefore P(x_0, y_0, z_0) = (1, -1, 1)$$

94. a) If  $z = e^{xy}$ ,  $x = 2u + v$ ,  $y = \frac{u}{v}$  find

b) Locate all relative extrema of  $f(x, y) = x^2 + y^2 - 8y$ .

$$= e^{(2u+v) \times \frac{u}{v}} \left[ \frac{2u}{v} + \frac{2u+v}{v} \right]$$

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b) Critical points

$$f_x = 6x - 2y$$

$$f_y = -2x + 2y - 8$$

$$f_x = 0$$

$$\Rightarrow 6x - 2y = 0 \longrightarrow (1)$$

$$f_y = 0$$

$$\Rightarrow -2x + 2y - 8 = 0 \longrightarrow (2)$$

$$(1) + (2)$$

$$= 4x - 8 = 0$$

$$\therefore x = 2$$

$$y = 6$$



$$D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - f_{xy}^2$$

$$f_{xx} = 6$$

$$f_{yy} = 2$$

$$f_{xy} = -2$$

$$\therefore D = (6 \times 2) - 4$$
$$= 8$$

$$D > 0$$

$$f_{xx} = 6 > 0$$

$\therefore F$  has relative minimum at

$$\begin{aligned} x &\longrightarrow 0 \text{ to } y \\ y &\longrightarrow 0 \text{ to } \infty \end{aligned}$$

∴ Order of Integration now is

$$\begin{aligned} &\int_0^{\infty} \int_0^y \frac{e^{-y}}{y} dx dy \\ &= \int_0^{\infty} \frac{e^{-y}}{y} [x]_0^y dy \quad \Rightarrow \quad \int_0^{\infty} \frac{e^{-y}}{\cancel{y}} [\cancel{y}] dy \\ &\Rightarrow \int_0^{\infty} e^{-y} dy \quad \Rightarrow \quad [e^{-y}]_0^{\infty} \quad \Rightarrow \quad [e^{-\infty}] \\ &\Rightarrow [0 + 1] = \underline{1}. \end{aligned}$$

$$x+2y=6$$

↳

|   |   |   |
|---|---|---|
| x | 0 | 6 |
| y | 3 | 0 |

The points are (0,3) (6,0)

$y \rightarrow 0$  to  $3-x/2$  ;  $x \rightarrow$

$$\int_0^6 \int_0^{3-\frac{x}{2}} \int_0^{6-2y-x} dz dy dx$$

$$\Rightarrow \int_0^6 \int_0^{3-\frac{x}{2}} [6-2y-x] dy dx$$

$$\Rightarrow \int_0^6 \left[ 6 \left[ y \right]_0^{3-\frac{x}{2}} - 2 \left[ \frac{y^2}{2} \right]_0^{3-\frac{x}{2}} - x \left[ y \right]_0^{3-\frac{x}{2}} \right] dx$$

$$\Rightarrow \int_0^6 \left[ 6 \left( 3-\frac{x}{2} \right) - x \left( 3-\frac{x}{2} \right) - \frac{2}{2} \left[ 9 - \frac{6x}{2} + \frac{x^2}{4} \right] \right] dx$$

$$\Rightarrow \int_0^6 \left[ 18 - \frac{6x}{2} - 3x + \frac{x^2}{2} - 9 + 3x - \frac{x^2}{4} \right] dx$$

$$\Rightarrow (9 \times 6) - (3 \times 18) + \left(\frac{1}{4} \times 72\right)$$

$$\Rightarrow 54 - 54 + 18$$

$\Rightarrow$

18

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We have  $\phi(x, y) = xy$

$$y \rightarrow 0 \text{ to } 1x$$

$$x \rightarrow 0 \text{ to } 1$$

$$M = \int_0^1 \int_0^{1-x} xy \, dy \, dx$$

$$= \int_0^1 x \left[ \frac{y^2}{2} \right]_0^{1-x} dx \Rightarrow \int_0^1 x \left[ \frac{1+x^2-2x}{2} \right] dx$$

$$\Rightarrow \int_0^1 \frac{1}{2} x [1+x^2-2x] dx \Rightarrow \frac{1}{2} \int_0^1 x+x^3-2x^2 dx$$

$$\Rightarrow \frac{1}{2} \left[ \left[ \frac{x^2}{2} \right]_0^1 + \left[ \frac{x^4}{4} \right]_0^1 - 2 \left[ \frac{x^3}{3} \right]_0^1 \right] \Rightarrow \frac{1}{2} \left[ \frac{1}{2} + \frac{1}{4} - \frac{2}{3} \right]$$

$$\Rightarrow \frac{1}{2} \left[ \frac{1}{12} \right] \Rightarrow \frac{1}{24}$$

$$\text{Centre of gravity} = \bar{x} = \frac{1}{M} \cdot \iint_R x \phi(x, y) \, dA$$

$$\text{We have } M = \frac{1}{24},$$



$$\Rightarrow 12 \int_0^1 (x^2 + x^4 - 2x^3) dx$$

$$\Rightarrow 12 \left[ \left[ \frac{x^3}{3} \right]_0^1 + \left[ \frac{x^5}{5} \right]_0^1 - \left[ \frac{2x^4}{4} \right]_0^1 \right]$$

$$\Rightarrow 12 \left[ \frac{1}{3} + \frac{1}{5} - \frac{2}{4} \right] \Rightarrow 12 \times \left[ -\frac{1}{30} \right]$$

$$\Rightarrow \frac{12}{30} = \frac{2}{5}$$

$$\frac{2}{5}$$

Centre of gravity  $\bar{y} = \frac{1}{M} \iint_R y \delta(x,y) dA$

$$= 24 \int_0^1 \int_0^{1-x} xy^2 dy dx$$

$$\Rightarrow 24 \int_0^1 x \left[ \frac{y^3}{3} \right]_0^{1-x} dx$$

$$\Rightarrow 24 \int_0^1 x \left[ \frac{1 - 3x + 3x^2 - x^3}{3} \right] dx$$

$$\Rightarrow \frac{24}{3} \int_0^1 x (1 - 3x + 3x^2 - x^3) dx$$

$$\Rightarrow 8 \int_0^1 (x - 3x^2 + 3x^3 - x^4) dx$$

$$\Rightarrow 8 \left[ \left[ \frac{x^2}{2} \right]_0^1 - 3 \left[ \frac{x^3}{3} \right]_0^1 + 3 \left[ \frac{x^4}{4} \right]_0^1 - \left[ \frac{x^5}{5} \right]_0^1 \right] \Rightarrow 8 \left[ \frac{1}{2} - 1 + \frac{3}{4} - \frac{1}{5} \right]$$

$$\Rightarrow 8 \left[ \frac{1}{2} - 1 + \frac{3}{4} - \frac{1}{5} \right]$$

$y=x$  and  $y=x^2$

$$\text{ans:- } \int_0^1 \int_{x^2}^x x^2 dy dx$$

$$= \int_0^1 x^2 [y]_{x^2}^x dx$$

$$= \int_0^1 x^2 (x - x^2) dx$$

$$= \left[ \frac{x^4}{4} \right]_0^1 - \left[ \frac{x^5}{5} \right]_0^1$$

$$= \frac{1}{4} - \frac{1}{5} = \underline{\underline{\frac{1}{20}}}$$

$$= a \cdot \frac{a^7}{2} + \frac{a^8}{2} + a \cdot \frac{a^7}{2} \cdot \frac{a^7}{2} + a \cdot \frac{a^7}{2} \cdot \frac{a^7}{2}$$

6. Use Polar co-ordinates to evaluate  $\int_{-1}^1$

ans:-  $x = -1, x = 1, y = 0, y = \sqrt{1-x^2} \Rightarrow x^2 + y^2 = 1$

Substitute  $x = r \cos \theta, y = r \sin \theta, x^2 + y^2 = r^2$

$$dy dx = r dr d\theta.$$

$$\int_{-1}^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2)^{3/2} = \int_0^1 \int_0^{\pi} (r^2)^{3/2} \cdot r d\theta dr$$

$$= \int_0^1 \int_0^{\pi} r^4 d\theta dr = \int_0^1 r^4 [0]_0^{\pi} dr$$

$$= \pi \cdot \left[ \frac{r^5}{5} \right]_0^1 = \frac{\pi}{5}$$

$$1 - \lim_{k \rightarrow \infty} \frac{a_k}{b_k} = \lim_{k \rightarrow \infty} \frac{1}{(2k+3)^{17}}$$

$$= \lim_{k \rightarrow \infty} \frac{k^{17}}{k^{17} \left(2 + \frac{3}{k}\right)^{17}}$$

$$= \frac{1}{2} > 0 \neq \text{finite}$$

by limit comparison test, since  $\sum_{k=1}^{\infty} \frac{1}{(2k+3)^{17}}$

$\sum_{k=1}^{\infty} \frac{1}{(2k+3)^{17}}$  also converges.

8. Examine whether the series converges

$$\sum_{k=1}^{\infty} \frac{1}{(\ln(k+1))^k}$$

Ans: By root test,  $\rho = \lim_{k \rightarrow \infty} (a_k)^{1/k}$

$$= \lim_{k \rightarrow \infty} \left( \frac{1}{(\ln(k+1))^k} \right)^{1/k}$$

$$= \lim_{k \rightarrow \infty} \frac{1}{\ln(k+1)}$$

$\therefore$  the series converges.

$$\begin{aligned}
 \lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} &= \lim_{k \rightarrow \infty} \frac{(k+1)!}{(k+1)^{k+1}} \cdot \frac{k^k}{k!} \\
 &= \lim_{k \rightarrow \infty} \frac{(k+1) \cdot k^k}{(k+1)^k \cdot (k+1)} \\
 &= \lim_{k \rightarrow \infty} \left( \frac{k}{k+1} \right)^k = \lim_{k \rightarrow \infty} \left( \frac{k}{k+1} \right)^k
 \end{aligned}$$

$\therefore$  by ratio test  $\sum a_k$  converges

i) Let  $a_k = \left( \frac{k}{k+1} \right)^{k^2}$

$$\begin{aligned}
 \lim_{k \rightarrow \infty} a_k &= \lim_{k \rightarrow \infty} \left[ \left( \frac{k}{k+1} \right)^{k^2} \right]^{\frac{1}{k}} = \lim_{k \rightarrow \infty} \left( \frac{k}{k+1} \right)^k \\
 &= \lim_{k \rightarrow \infty} \left( \frac{1}{1 + \frac{1}{k}} \right)^k = \lim_{k \rightarrow \infty} \left( \frac{1}{1 + \frac{1}{k}} \right)^k
 \end{aligned}$$

$\therefore$  by root test the series converges



$$\begin{aligned} \lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} &= \lim_{k \rightarrow \infty} \frac{x^{k+1}}{(2k+1)(2k+3)} \cdot (2k) \\ &= \lim_{k \rightarrow \infty} \frac{x(2k-1)}{(2k+3)} = \end{aligned}$$

If  $x < 1$ , the series converges, and if  $x > 1$ , the series diverges:

If  $x = 1$ , the series becomes, a p-series

$$\text{Let } a_k = \frac{1}{(2k-1)(2k+1)} = \frac{1}{4k^2 - 1}$$

$$\text{Let } b_k = \frac{1}{k^2}$$

$$\lim_{k \rightarrow \infty} \frac{a_k}{b_k} = \lim_{k \rightarrow \infty} \frac{k^2}{4k^2 - 1} = \frac{1}{4}$$

$\therefore$  by limit comparison test, since  $\sum b_k$  is a convergent p-series  $\sum a_k$  is also convergent.

$$\lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} = \lim_{k \rightarrow \infty} \frac{1 \cdot 3 \cdot 5 \cdots (2k+1)}{(2k+1)!}$$

$$= \lim_{k \rightarrow \infty} \frac{(2k+1)}{2k(2k+1)} = 0 < 1$$

$\therefore$  the series converges by ratio test.

18 b) Prove that the series  $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{k}{k!}$  is conditionally convergent.

Ans: Let  $a_k = \frac{k-2}{k(k+1)}$

$$a_{k+1} = \frac{(k+1)-2}{(k+1)(k+2)} = \frac{k-1}{(k+1)(k+2)}$$

$$\therefore a_k > a_{k+1}$$

$$\lim_{k \rightarrow \infty} a_k = \lim_{k \rightarrow \infty} \frac{k-2}{k^2(1+\frac{1}{k})} = 0$$

$\therefore$  by Leibnitz test the series converges.

$$= 1 > 0$$

Since  $\sum b_k$  is a divergent p-limit comparison test  $\sum |a_k|$  is also

~~$\sum a_k$  converges &  $\sum |a_k|$  diverges~~

$\sum |a_k|$  diverges  $\Rightarrow \sum a_k$  diverges

Since  $\sum a_k$  converges & diverges as

$\Rightarrow \sum a_k$  is conditionally convergent

$$f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$$

$$\text{Given } f(x) = \frac{1}{x+1}$$

$$f(0) = \frac{1}{0+1} = 1$$

$$f'(x) = \frac{-1}{(1+x)^2}$$

$$f'(0) = \frac{-1}{1^2} = -1$$

$$f''(x) = \frac{+2}{(1+x)^3}$$

$$f''(0) = \frac{+2}{1^3} = +2$$

$$f'''(x) = +2 \times -3 \times (1+x)^{-4}$$

$$= \frac{-6}{(1+x)^4}$$

$$f'''(0) = -6$$

$$\therefore f(x) = 1 + x \times -1 + \frac{x^2}{2} \times 2 + \frac{x^3}{6} \times (-6) + \dots$$

$$= 1 - x + x^2 - x^3 + \dots$$

=====

$$= \frac{2}{\pi} \int_0^{\pi} x \sin(nx) dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x \sin(nx) dx$$

$$= \frac{2}{\pi} \left[ x \times \frac{\cos(nx)}{n} - 1 \times \frac{-1}{n} \frac{\sin(nx)}{n} \right]$$

$$= \frac{2}{\pi} \left[ \frac{-\pi}{n} \cos(n\pi) + \frac{1}{n^2} \sin(n\pi) \right]$$

$$= \frac{2}{\pi} \left[ \frac{-\pi}{n} \times (-1)^n \right]$$

$$= -\frac{2}{n} (-1)^n$$

$$f(x) = \sum_{n=1}^{\infty} \underline{\underline{-\frac{2}{n} (-1)^n \sin(nx)}}$$



$$= \frac{1}{2L} \int_{-\pi}^{\pi} f(x) dx$$

$$= \frac{1}{2\pi} \times 2 \int_0^{\pi} |\sin x| dx$$

$$= \frac{1}{\pi} \int_0^{\pi} \sin x dx$$

$$= \frac{1}{\pi} [-\cos x]_0^{\pi}$$

$$= \frac{1}{\pi} [-\cos \pi - (-\cos 0)]$$

$$= \frac{1}{\pi} [-1 + 1] = \frac{2}{\pi}$$

$$a_n = \frac{1}{L} \int_{-\pi}^{\pi} f(x) \cos\left(\frac{n\pi x}{L}\right) dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} |\sin x| \cos(nx) dx$$

$$= \frac{2}{\pi} \int_0^{\pi} \sin x \cos(nx) dx$$

$$= \frac{1}{\pi} \int_0^{\pi} \sin(1+n)x + \sin(1-n)x dx$$

$$= \frac{1}{\pi} \int_0^{\pi} \sin(n+1)x - \sin(n-1)x dx$$

$$= \frac{1}{\pi} \left[ -\frac{\cos(n+1)x}{n+1} + \frac{\cos(n-1)x}{n-1} \right]_0^{\pi}$$

$$= -\frac{\cos 2x}{2} \Big|_0^{\pi}$$

$$= -\frac{1}{2} [\cos 2\pi - \cos 0]$$

$$= -\frac{1}{2} [1 - 1]$$

$$= \underline{\underline{0}}$$

series in  $(0, \pi)$ .

Fourier sine series of  $f(x)$  is,

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx$$
$$= \frac{2}{\pi} \left[ \int_0^{\pi/2} kx \sin nx \, dx + \int_{\pi/2}^{\pi} k(\pi - x) \sin nx \, dx \right]$$

$$= \frac{2}{\pi} \left\{ \left[ 1/x \left[ -\frac{\cos nx}{n} \right] - k \left[ -\frac{\sin nx}{n} \right] \right] \right.$$
$$\left. + \left[ k(\pi - x) \left( -\frac{\cos nx}{n} \right) - \left( -\frac{\sin nx}{n} \right) \right] \right\}$$

$$= \frac{2k}{\pi} \left\{ \left[ \frac{-\frac{\pi}{2} \cos \frac{n\pi}{2}}{n} + \frac{1}{n^2} \sin \frac{n\pi}{2} \right] - \left[ 0 \right] \right\}$$

20 a) Find the Fourier cosine series of  $(0, \pi)$ . Hence show that  $1 + \frac{1}{2^2} + \frac{1}{3^2}$

Fourier cosine series of  $f(x)$  is

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right)$$

$$a_0 = \frac{1}{l} \int_0^l f(x) dx$$

$$= \frac{1}{\pi} \int_0^{\pi} x^2 dx$$

$$= \frac{1}{\pi} \left[ \frac{x^3}{3} \right]_0^{\pi} = \frac{1}{\pi} \times \frac{\pi^3}{3} = \frac{\pi^2}{3}$$

$$a_n = \frac{2}{l} \int_0^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x^2 \cos(nx) dx$$

$$+ \frac{4}{3^2} x - 1 \times \cos 3x$$

$$x^2 = \frac{\pi^2}{3} + 4 \left[ -\cos x + \frac{1}{2^2} \cos 2x \right]$$

When  $x = \pi$ ,

$$\pi^2 = \frac{\pi^2}{3} + 4 \left[ -\cos \pi + \frac{1}{2^2} \cos 2\pi \right]$$

$$\pi^2 = \frac{\pi^2}{3} + 4 \left[ 1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right]$$

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{1}{4} \left[ \pi^2 - \frac{\pi^2}{3} \right]$$

$$= \frac{1}{4} \left[ \frac{2\pi^2}{3} \right]$$

$$= \frac{\pi^2}{6}$$

$$\therefore 1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots =$$





$$= \frac{1}{2} \left[ \left[ \frac{x^2}{2} \right]_0^1 + \left[ x - \frac{x^2}{2} \right]_1^2 \right]$$

$$= \frac{1}{2} \left[ \frac{1}{2} + 2 - \frac{4}{2} - \left( 1 - \frac{1}{2} \right) \right]$$

$$= \frac{1}{2} \left[ \frac{1}{2} - 1 + \frac{1}{2} \right]$$

$$= 0$$

$$a_n = \frac{1}{l} \int_a^{a+2l} f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$= \int_0^1 x \cos(n\pi x) dx + \int_1^2 (1-x) \cos(n\pi x) dx$$

$$= \left[ x \times \frac{\sin(n\pi x)}{n\pi} - 1 \times \frac{1}{n^2 \pi^2} x^{-\cos(n\pi x)} \right]_0^1 +$$

$$- 1 \times \frac{1}{n^2 \pi^2} \cos(n\pi x)$$

$$= \frac{1}{n\pi} \times \sin n\pi + \frac{1}{n^2 \pi^2} \cos n\pi - \left( \frac{1}{n^2 \pi^2} \right) + \frac{1}{n^2 \pi^2}$$

$$= \frac{1}{n^2 \pi^2} (-1)^n - \frac{1}{n^2 \pi^2} - \frac{1}{n^2 \pi^2}$$

0

$$= x \times \frac{-\cos(n\pi x)}{n\pi} - 1 \times \frac{-1}{n\pi} \times \sin(n\pi x)$$

$$+ (1-x) \cdot \frac{-\cos(n\pi x)}{n\pi} - -1 \times \frac{-1}{n^2\pi^2} \sin(n\pi x)$$

$$= \frac{-1}{n\pi} (-1)^n + \frac{1}{n\pi} \times 1$$

$$= \frac{1}{n\pi} [1 - (-1)^n]$$

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$$f(x) = \frac{2}{n^2\pi^2} (-1)^{n-1} \cos(n\pi x) + \frac{1}{n}$$

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